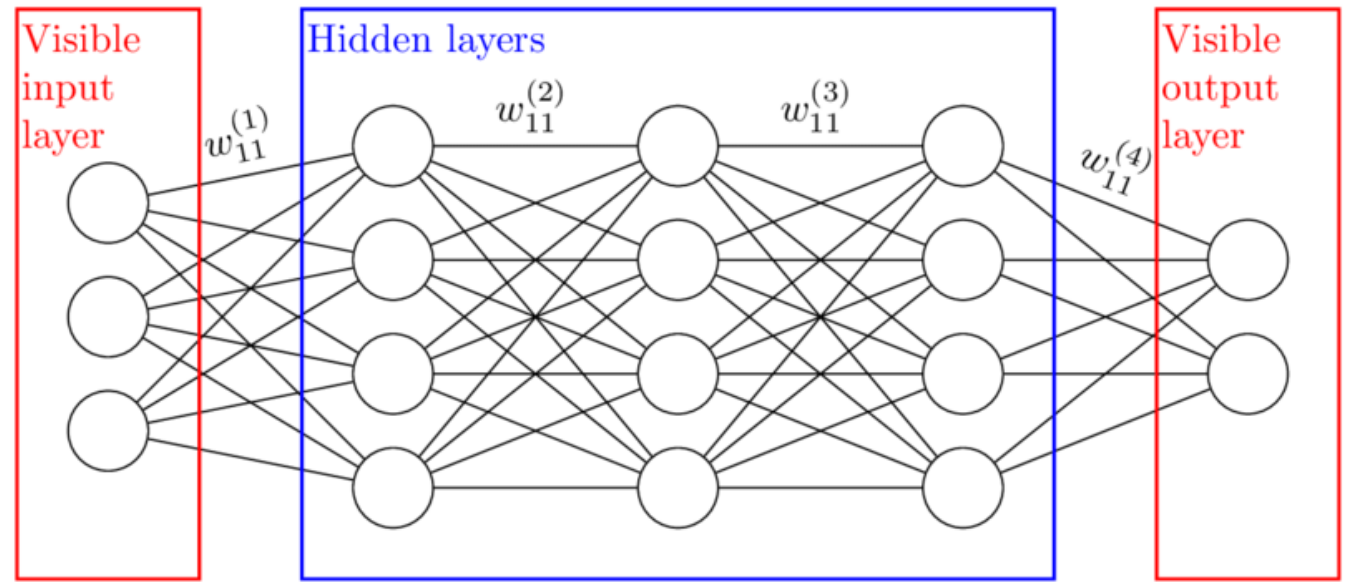


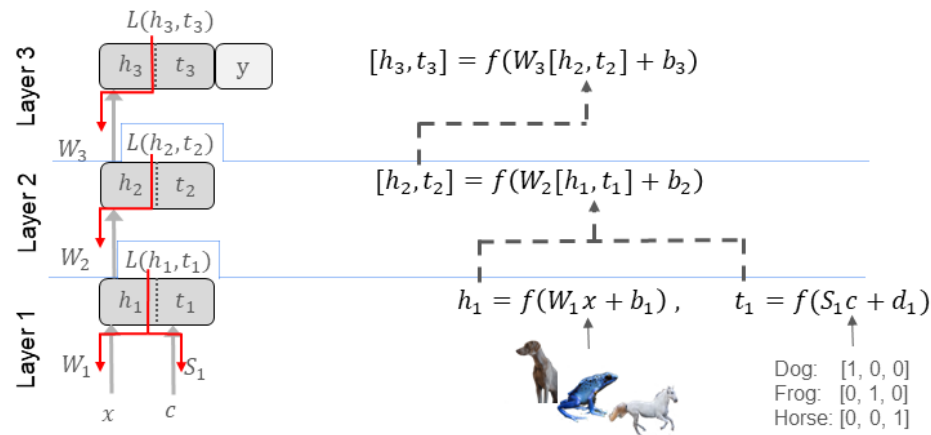
Designing the Network: Input, Hidden, and Output Layers



- **Input Layer:** Receives the raw data (e.g., 784 pixels for a 28×28 image).
- **Hidden Layers:** Intermediate layers where the "learning" happens. They extract increasingly complex features from the data.
- **Output Layer:** Produces the final prediction, such as a category or a probability.

The Forward Pass: From Data to Decision

Complete Forward Pass with Learning and Inference



- Data flows in one direction: from input to output.
- Each layer calculates $z = W \cdot x + b$, then applies an activation function.
- The final output is the network's "guess" or raw score (logit).

Weight Initialization: Setting the Stage for Success

Weight Initialization

$$W^{[l]} = \begin{bmatrix} w_{11}^{[l]} & w_{12}^{[l]} & \dots & w_{1n^{[l-1]}}^{[l]} \\ w_{21}^{[l]} & w_{22}^{[l]} & \dots & w_{2n^{[l-1]}}^{[l]} \\ \vdots & \vdots & \dots & \vdots \\ w_{n^{[l]}1}^{[l]} & w_{n^{[l]}2}^{[l]} & \dots & w_{n^{[l]}n^{[l-1]}}^{[l]} \end{bmatrix}$$



- Initializing weights to zero prevents learning because neurons remain symmetric.
- **Xavier (Glorot) Initialization:** Best for *Sigmoid* or *Tanh* activations; it keeps the variance of the signals constant across layers.
- **He Initialization:** *Optimized* for *ReLU* activations; it uses a larger variance ($\sqrt{2}/n$) to prevent the signal from dying out in deep networks.